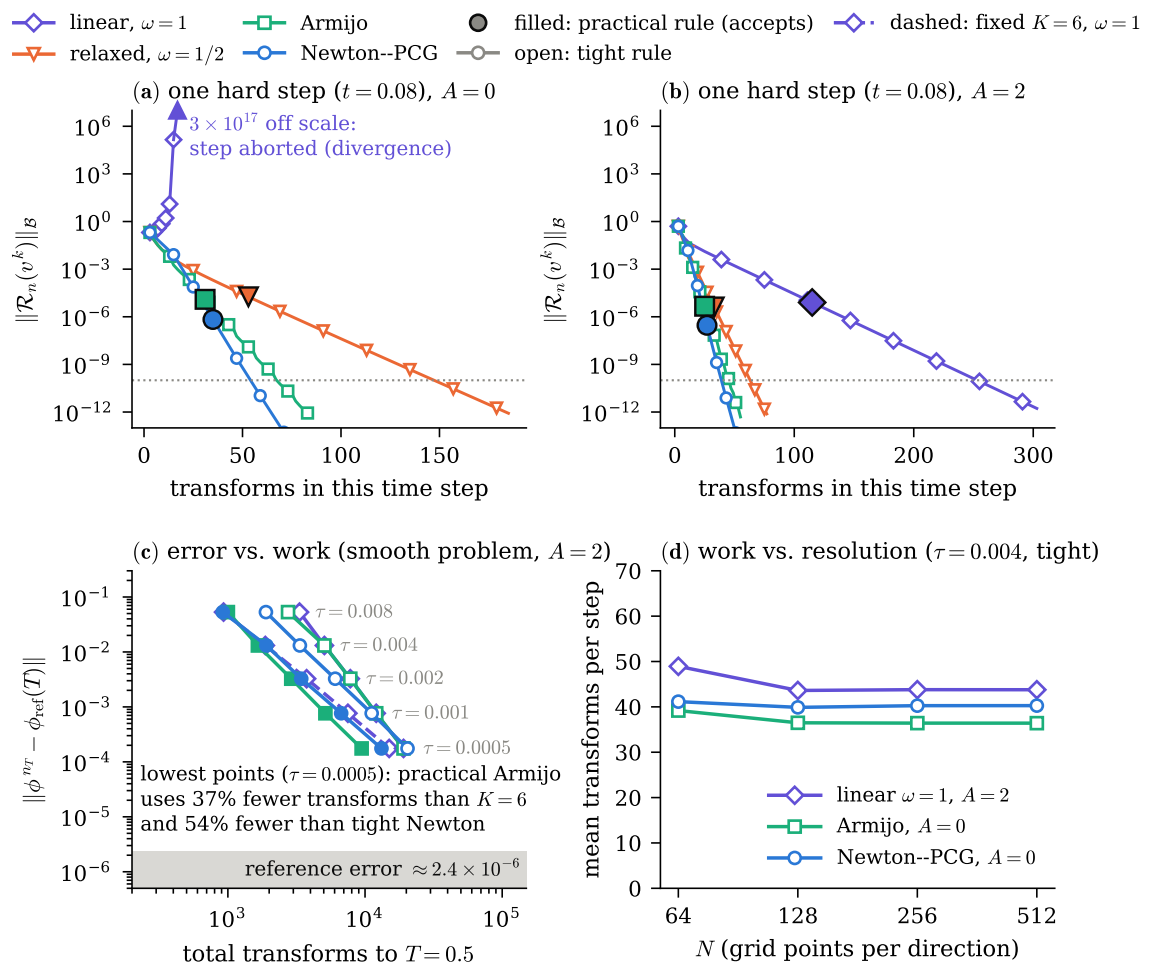

linear, $\omega = 1$
relaxed, $\omega = 1/2$
Armijo
Newton--PCG
filled: practical rule (accepts)
open: tight rule
dashed: fixed $K = 6$, $\omega = 1$
(a) one hard step ($t = 0.08$), $A = 0$
3×10^{17} off scale:
step aborted (divergence)
$\|\mathcal{R}_n(v^k)\|_{\mathcal{B}}$
transforms in this time step
(b) one hard step ($t = 0.08$), $A = 2$
$\|\mathcal{R}_n(v^k)\|_{\mathcal{B}}$
transforms in this time step
(c) error vs. work (smooth problem, $A = 2$)
$\|\phi^{n_T} - \phi_{\mathrm{ref}}(T)\|$
$\tau = 0.008$
$\tau = 0.004$
$\tau = 0.002$
$\tau = 0.001$
$\tau = 0.0005$
lowest points ($\tau = 0.0005$): practical Armijo
uses 37% fewer transforms than $K = 6$
and 54% fewer than tight Newton
reference error $\approx 2.4 \times 10^{-6}$
total transforms to $T = 0.5$
(d) work vs. resolution ($\tau = 0.004$, tight)
mean transforms per step
linear $\omega = 1$, $A = 2$
Armijo, $A = 0$
Newton--PCG, $A = 0$
N (grid points per direction)